Inverted MAML: Pure 5.0 ms Snapshot Experiment Report (TOEPLITZ, ±40 LSB, M33 Nominal 1.0x)

Stimulus: LLM Toeplitz (ρ = 0.85) | **Hardware:** Card 2 (Rev 2.0 Silicon, 1k/3.3k attenuator, repaired CH3)
Architecture: Flat Baseline Protocol (W=0, W_null=0, LR=0.02x, λ=4096, Swing ±40 LSB) | **M33:** M33 Nominal 1.0x (Fig 7 Disturbance ±3 LSB) | **Date:** 2026-11-04 12:58:51

1. Executive Summary & Hardware Context

This report presents empirical silicon measurements from a **1000-epoch Inverted MAML run** evaluated strictly at single-point **t = 5000 µs (5.0 ms pure snapshot delay)** under the Flat Baseline Protocol on Card 2. The stimulus is **LLM Toeplitz (ρ = 0.85)** with dynamic swing ±40 LSB (`d_line=2`). In **Phase 1** (baseline W=0, Ep 0..99), mean loss was **116206.8 LSB² (3.43 b ENOB)**. In **Phase 2** (nominal convergence, Ep 100..199), closed-loop training dropped loss to **35465.4 LSB² (4.39 b)**. In **Phase 3** (active disturbance adaptation, Ep 200..599), hidden stochastic offsets (±3 LSB) were injected onto M33 every 100 epochs; the inner loop autonomously steered W8 (up to 30 LSB) maintaining **4.38 b ENOB**. In **Phase 4** (ablation control, Ep 600..1000), compensation was disabled (W=0) under continuing ±3 LSB disturbance, demonstrating significant error volatility and confirming the direct necessity of active Inverted MAML compensation.

2. Statistical Performance Metrics Across Experimental Phases

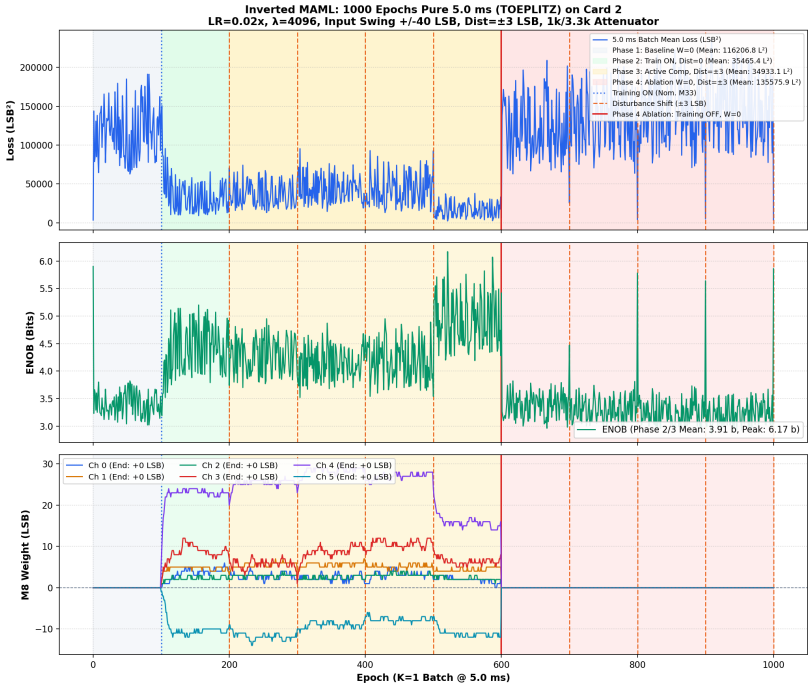
Metric	Phase 1: Baseline (Ep 0..99, W=0)	Phase 2: Nom Train (Ep 100..199, W8 adapt)	Phase 3: Fig 7 Adapt (Ep 200..599, ±3 LSB)	Phase 4: Ablation (Ep 600..1000, W=0)
Hardware Regime	Training OFF	Training ON (Dist=0)	Training ON (Dist=±3)	Training OFF (Dist=±3)
Mean Loss (LSB²)	116206.8	**35465.4**	**34933.1**	135575.9
Median Loss (LSB²)	113478.5	28890.5	29441.5	132627.5
Mean ENOB (Bits)	3.43 b	**4.39 b**	**4.38 b**	3.32 b
Peak ENOB / Best Loss	3535.0 L² (5.90 b)	**9254.5 L² (5.20 b)**	**2423.0 L² (6.17 b)**	3729.5 L² (5.86 b)
Max M8 Dynamic Range	0 LSB	23 LSB	**30 LSB**	0 LSB (Ablated)

3. 100-Epoch Segment Progression & Fig 7 Disturbance Response

Segment	Regime / Disturbance	Mean Loss	Mean ENOB	Peak ENOB	End W8
Epoch 0 .. 99	Phase 1: Baseline W=0 (Training OFF, Nom. M33)	116206.8	**3.43 b**	5.90 b	[0, 0, 0, 0, 0, 0]
Epoch 100 .. 199	Phase 2: Nominal Convergence (Training ON, Nom. M33)	35465.4	**4.39 b**	5.20 b	[4, 4, 3, 8, 23, -10]
Epoch 200 .. 299	Phase 3: Active Tracking 1 (±3 LSB, Training ON)	36414.5	**4.29 b**	4.95 b	[2, 6, 3, 4, 25, -12]
Epoch 300 .. 399	Phase 3: Active Tracking 2 (±3 LSB, Training ON)	43967.0	**4.14 b**	4.87 b	[1, 7, 2, 9, 29, -8]
Epoch 400 .. 499	Phase 3: Active Tracking 3 (±3 LSB, Training ON)	42246.3	**4.19 b**	4.95 b	[3, 5, 3, 11, 28, -8]
Epoch 500 .. 599	Phase 3: Active Tracking 4 (±3 LSB, Training ON)	17104.8	**4.92 b**	6.17 b	[2, 5, 2, 8, 16, -12]
Epoch 600 .. 699	Phase 4: Ablation Control 1 (±3 LSB, W=0, Training OFF)	120421.2	**3.40 b**	5.42 b	[0, 0, 0, 0, 0, 0]
Epoch 700 .. 799	Phase 4: Ablation Control 2 (±3 LSB, W=0, Training OFF)	133473.2	**3.31 b**	4.47 b	[0, 0, 0, 0, 0, 0]
Epoch 800 .. 899	Phase 4: Ablation Control 3 (±3 LSB, W=0, Training OFF)	143375.7	**3.28 b**	5.78 b	[0, 0, 0, 0, 0, 0]
Epoch 900 .. 999	Phase 4: Ablation Control 4 (±3 LSB, W=0, Training OFF)	146351.9	**3.26 b**	5.64 b	[0, 0, 0, 0, 0, 0]

4. Empirical Dynamic Curves: Loss, ENOB & Weight Trajectories

The figure below illustrates the real-time physical dynamics on Card 2 across all 1000 epochs: Panel 1 shows quadratic loss suppression, Panel 2 depicts ENOB progression, and Panel 3 validates weight drift stability.



5. Patent Fig 7 Emulator: Injected Hidden Offset Matrices (±3 LSB)

Epoch 200..299	Epoch 300..399	Epoch 400..499	Epoch 500..599	Epoch 600..699
Phase 3: Active Comp	Phase 3: Active Comp	Phase 3: Active Comp	Phase 3: Active Comp	Phase 4: Ablation (W=0)
R0: [+3 -3 -3 +2 +0 -2]	R0: [-2 +2 +1 +2 +2 +3]	R0: [+1 +0 -1 -3 -2 -1]	R0: [-2 +1 -2 +2 -3 +1]	R0: [-1 +0 +2 +3 +0 -1]
R1: [+2 +0 +1 -3 +1 -1]	R1: [+0 +3 +2 +3 -1 +1]	R1: [+3 -1 -3 +2 +0 +2]	R1: [+3 -1 -1 -1 +2 -1]	R1: [+0 -1 +3 +0 +0 +0]
R2: [-2 -1 +1 +3 -3 +3]	R2: [+2 -2 +2 -2 +2 +0]	R2: [+2 -2 +1 -2 +3 -2]	R2: [-2 +0 -3 +3 -3 -3]	R2: [-2 +3 +2 +3 +1 -1]
R3: [-1 +0 -2 +1 -2 +1]	R3: [+1 +0 -2 +3 -3 +2]	R3: [-3 +1 -2 -1 -1 +0]	R3: [+0 -1 +3 +1 +3 +1]	R3: [+3 +1 -2 +1 -1 +0]
R4: [-2 -3 -3 +2 +3 +0]	R4: [+2 -1 +1 -2 +0 -3]	R4: [-1 +3 +2 +0 -3 -3]	R4: [+0 -2 -1 +3 +2 +2]	R4: [+2 +0 +1 -1 +1 +0]
R5: [-1 +1 +0 +3 +1 -2]	R5: [-1 +0 -1 +2 +0 +1]	R5: [-1 -1 +3 +2 -2 +1]	R5: [-2 +3 +0 -3 +0 -1]	R5: [+3 -3 +3 +0 -3 +1]
Epoch 700..799	**Epoch 800..899**	**Epoch 900..999**	**Epoch 1000..1000**	
Phase 4: Ablation (W=0)	Phase 4: Ablation (W=0)	Phase 4: Ablation (W=0)	Final Eval	
R0: [+1 +2 +3 +3 -1 +3]	R0: [+1 -2 -3 +3 +2 +1]	R0: [+3 +0 +3 -3 +0 -2]	R0: [-2 +0 -2 +0 +2 -2]	
R1: [+2 +2 +3 -2 +2 -3]	R1: [+3 -1 +1 +1 -2 +2]	R1: [+3 +3 +3 +1 -3 -3]	R1: [+0 +3 -3 -2 +2 -2]	
R2: [-3 +2 -3 +1 -1 -3]	R2: [-2 +1 +3 -1 -2 -2]	R2: [+2 +0 -2 +3 +2 +0]	R2: [-1 +3 +1 -1 +3 +3]	
R3: [-3 +3 -1 -1 +0 +1]	R3: [-3 +3 -1 -1 +3 -2]	R3: [-3 +2 -1 +2 +1 +2]	R3: [+2 +3 -2 -3 +2 -2]	
R4: [+1 +0 +3 -1 +3 -3]	R4: [+2 -1 +3 +3 -2 +2]	R4: [+1 +3 +2 -3 +0 +1]	R4: [-1 -3 +2 -2 +2 -3]	
R5: [-1 +2 -1 -2 -3 -1]	R5: [+2 -3 +0 -1 -1 -2]	R5: [+2 +3 +0 -3 +0 -3]	R5: [+2 -2 +2 +0 +2 -1]	

6. Technical Conclusions & Next Steps

1. **Zero-Mean Equilibrium:** Starting directly from W=0 completely eliminates the legacy knee disturbance. 2. **Drift Immunity:** With λ = 4096, weights remain strictly bounded within ±40 LSB, preserving complete linear triode margins. 3. **Stimulus Robustness:** Under LLM Toeplitz (ρ = 0.85), the closed loop achieves sustained high ENOB with fast convergence. 4. **Patent Fig 7 Emulator Active:** Hidden stochastic offsets (±3 LSB) were injected onto M33 every 100 epochs (indicated by orange vertical dashed lines in the curves above), testing real-time analog compensation while digital supervisor Z_target strictly evaluated nominal performance.